

REAL-TIME DATA LAKE PIPELINE FOR E-COMMERCE ANALYTICS

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Problem Statement

- Modern e-commerce platforms generate large volumes of real-time user interaction data (clicks,purchases,searches).
- Traditional databases struggle to process and store streaming data efficiently.
- Businesses require real-time analytics to understand customer behavior and improve decision-making.

Objectives

- Design a scalable real-time data pipeline for e-commerce events.
- Capture user interactions such as product views, clicks, and purchases.
- Process streaming data using distributed computing.
- Store processed data in a data lake for analytics and future ML applications.
- Demonstrate the end-to-end flow of data from Frontend to data lake storage.

Pipeline

- Frontend (React)Captures user interactions on the website.
- Backend API (Express.js)Receives event data from the frontend.
- Event Streaming (Kafka)Streams user events in real time.
- Stream Processing (Apache Spark)Consumes Kafka events and processes them.
- Data Lake Storage (MinIO)Stores processed data in scalable object storage.
- Pipeline Flow:
Frontend → Express API → Kafka → Spark → MinIO Data Lake

Tech Stack

- Frontend: React
- Backend: Express.js / Node.js
- Database: MongoDB
- Event Streaming: Apache Kafka
- Processing Engine: Apache Spark
- Data Lake Storage: MinIO

Pipeline



System Output

- Real-time event data ingestion from the web application.
- Stream processing of user activity data.
- Structured event data stored in the MinIO data lake.
- Data ready for analytics, dashboards, and machine learning applications.

Results and Performance

- Successfully implemented real-time event ingestion pipeline.
- Pipeline handled continuous streaming of user interaction events.
- Achieved low-latency event processing using Kafka and Spark.
- Data stored efficiently in data lake format for future analytics.

Conclusion

- Built a complete real-time data lake pipeline for e-commerce systems.
- Demonstrated integration of modern big-data technologies.
- Pipeline enables scalable storage and analytics of streaming data.
- Architecture can support future applications like recommendation systems, analytics dashboards, and machine learning models.